

## KARLSBORG, Sweden

WMO: 025440

Lat: 58.517N

Lon: 14.517E

Elev: 308

StdP: 14.53

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |             | Humidification DP/MCDB and HR |            |            |            |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |           | WSF        |
|---------------|------------|-------------|-------------------------------|------------|------------|------------|------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|-----------|------------|
|               |            |             | 99.6%                         |            |            | 99%        |            |             | 0.4%                  |             | 1%          |             |                       |           |            |
|               | 99.6%      | 99%         | DP                            | HR         | MCDB       | DP         | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD      |            |
| (a)<br>1      | (b)<br>8.5 | (c)<br>12.9 | (d)<br>5.0                    | (e)<br>7.2 | (f)<br>9.8 | (g)<br>9.6 | (h)<br>9.1 | (i)<br>14.8 | (j)<br>26.5           | (k)<br>40.1 | (l)<br>23.4 | (m)<br>38.5 | (n)<br>5.5            | (o)<br>50 | (p)<br>N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.4                   | 78.3            | 65.4 | 75.1 | 63.6 | 71.8 | 62.6 | 67.9                | 74.5 | 65.9 | 72.3 | 64.1 | 69.6 | 6.5                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 65.6                            | 95.4 | 70.5 | 63.7 | 89.2 | 68.6 | 61.9 | 83.6 | 66.8 | 32.3          | 74.8 | 30.8 | 72.3 | 29.5 | 69.7 | 76.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 22.9              | 20.1 | 17.8 | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |
|                   |      |      | 5.2                        | 69.3 | 9.2                | 2.3 | -1.4                                               | 71.0 | -6.7       | 72.3 | -11.9      | 73.6 | -18.5      | 75.2 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 44.9        | 28.7  | 29.4  | 34.4  | 42.5  | 51.3  | 58.0  | 62.9  | 60.0  | 53.9  | 45.4  | 39.0  | 32.6  |
|                                                                        | DBStd       | 13.28       | 7.75  | 8.24  | 6.95  | 5.66  | 6.39  | 4.71  | 4.92  | 4.30  | 4.55  | 5.44  | 6.13  | 8.25  |
|                                                                        | HDD50       | 3062        | 660   | 576   | 484   | 232   | 60    | 1     | 0     | 0     | 15    | 161   | 333   | 540   |
|                                                                        | HDD65       | 7380        | 1125  | 996   | 949   | 675   | 424   | 214   | 105   | 167   | 332   | 607   | 781   | 1005  |
|                                                                        | CDD50       | 1213        | 0     | 0     | 1     | 8     | 101   | 242   | 398   | 309   | 134   | 19    | 1     | 0     |
|                                                                        | CDD65       | 54          | 0     | 0     | 0     | 0     | 0     | 5     | 38    | 11    | 0     | 0     | 0     | 0     |
|                                                                        | CDH74       | 400         | 0     | 0     | 0     | 1     | 15    | 68    | 264   | 52    | 0     | 0     | 0     | 0     |
|                                                                        | CDH80       | 51          | 0     | 0     | 0     | 0     | 0     | 4     | 39    | 8     | 0     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 8.4         | 9.0   | 9.1   | 8.9   | 7.8   | 7.5   | 7.6   | 7.2   | 7.3   | 8.3   | 9.3   | 9.7   | 9.4   |
| (15) Precipitation                                                     | PrecAvg     | 28.0        | 1.9   | 1.7   | 1.3   | 1.7   | 2.1   | 3.3   | 3.2   | 3.1   | 2.5   | 2.5   | 2.4   | 2.1   |
|                                                                        | PrecMax     | 34.5        | 3.6   | 3.2   | 2.2   | 4.0   | 4.4   | 6.1   | 6.8   | 5.1   | 5.5   | 5.3   | 4.6   | 4.3   |
|                                                                        | PrecMin     | 21.7        | 0.4   | 0.3   | 0.2   | 0.4   | 0.6   | 0.2   | 0.8   | 0.6   | 0.4   | 0.3   | 0.7   | 0.6   |
|                                                                        | PrecStd     | 3.7         | 0.9   | 0.7   | 0.5   | 0.9   | 0.9   | 1.4   | 1.4   | 1.3   | 1.6   | 1.2   | 1.1   | 0.9   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 45.3  | 48.2  | 56.8  | 64.7  | 76.1  | 79.8  | 84.5  | 78.8  | 70.5  | 60.3  | 53.1  | 50.8  |
|                                                                        |             | MCWB        | 43.8  | 43.5  | 48.9  | 51.0  | 59.6  | 63.2  | 66.9  | 66.0  | 63.1  | 56.5  | 51.9  | 49.4  |
|                                                                        | 2%          | DB          | 42.1  | 44.1  | 50.6  | 60.6  | 71.9  | 75.7  | 79.9  | 74.4  | 66.5  | 57.7  | 51.2  | 48.7  |
|                                                                        |             | MCWB        | 40.5  | 40.5  | 44.5  | 50.1  | 57.8  | 62.1  | 66.4  | 64.9  | 60.7  | 54.7  | 50.0  | 47.2  |
|                                                                        | 5%          | DB          | 40.2  | 41.9  | 46.9  | 57.1  | 67.5  | 72.0  | 76.7  | 70.6  | 63.8  | 55.7  | 49.0  | 45.4  |
|                                                                        |             | MCWB        | 38.6  | 39.8  | 42.1  | 48.2  | 57.2  | 60.6  | 65.4  | 63.1  | 58.6  | 53.2  | 47.8  | 43.6  |
|                                                                        | 10%         | DB          | 38.2  | 40.1  | 43.8  | 53.1  | 63.4  | 68.3  | 73.2  | 68.1  | 61.4  | 53.7  | 47.1  | 42.4  |
|                                                                        |             | MCWB        | 36.8  | 38.3  | 40.2  | 45.7  | 54.4  | 59.0  | 64.6  | 62.1  | 57.1  | 51.5  | 45.9  | 40.7  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 44.0  | 44.3  | 49.3  | 54.5  | 63.0  | 66.3  | 70.5  | 69.2  | 63.9  | 57.6  | 52.5  | 50.0  |
|                                                                        |             | MCDB        | 44.7  | 46.3  | 55.7  | 62.5  | 71.0  | 73.7  | 77.4  | 75.3  | 68.7  | 59.2  | 53.0  | 50.4  |
|                                                                        | 2%          | WB          | 40.8  | 41.5  | 45.5  | 50.9  | 60.5  | 64.1  | 69.0  | 66.5  | 61.6  | 55.5  | 50.2  | 47.0  |
|                                                                        |             | MCDB        | 41.9  | 43.2  | 49.8  | 58.0  | 68.1  | 72.6  | 75.9  | 71.2  | 65.2  | 57.4  | 51.1  | 48.3  |
|                                                                        | 5%          | WB          | 38.7  | 40.1  | 42.8  | 48.9  | 57.9  | 62.0  | 67.2  | 64.6  | 59.7  | 53.7  | 48.0  | 43.8  |
|                                                                        |             | MCDB        | 39.9  | 41.7  | 46.1  | 55.4  | 65.2  | 69.9  | 74.3  | 69.0  | 62.5  | 55.1  | 48.7  | 45.1  |
|                                                                        | 10%         | WB          | 36.8  | 38.3  | 40.6  | 46.7  | 55.6  | 60.0  | 65.5  | 63.1  | 58.0  | 51.8  | 46.2  | 40.8  |
|                                                                        |             | MCDB        | 38.1  | 39.9  | 43.3  | 52.6  | 62.3  | 67.1  | 72.1  | 67.1  | 60.5  | 53.4  | 46.9  | 42.1  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 7.8   | 8.7   | 11.4  | 15.9  | 16.9  | 16.3  | 15.4  | 14.2  | 12.2  | 9.6   | 7.0   | 7.5   |
|                                                                        |             | MCDBR       | 9.2   | 10.2  | 15.0  | 22.5  | 23.7  | 22.6  | 21.0  | 18.1  | 15.6  | 10.7  | 8.4   | 9.2   |
|                                                                        | 5% WB       | MCWBR       | 8.7   | 7.9   | 10.6  | 13.3  | 12.9  | 12.1  | 10.0  | 10.4  | 10.3  | 8.5   | 8.0   | 9.3   |
|                                                                        |             | MCWBR       | 9.1   | 9.2   | 14.3  | 19.6  | 20.4  | 20.6  | 18.4  | 16.0  | 14.8  | 10.0  | 8.1   | 9.1   |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.276 | 0.313 | 0.334 | 0.372 | 0.369 | 0.358 | 0.381 | 0.373 | 0.343 | 0.333 | 0.302 | 0.261 |
|                                                                        |             | taud        | 2.228 | 2.293 | 2.359 | 2.326 | 2.398 | 2.449 | 2.403 | 2.422 | 2.493 | 2.435 | 2.303 | 2.165 |
|                                                                        | Ebn at Noon | Ebn at Noon | 177   | 226   | 257   | 264   | 274   | 279   | 269   | 262   | 253   | 218   | 165   | 140   |
|                                                                        |             | Edh at Noon | 15    | 22    | 28    | 34    | 34    | 33    | 33    | 30    | 24    | 20    | 14    | 11    |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 110   | 306   | 788   | 1267  | 1637  | 1830  | 1636  | 1299  | 880   | 399   | 142   | 76    |
|                                                                        | RadStd      | RadStd      | 16    | 50    | 89    | 142   | 143   | 87    | 192   | 138   | 121   | 65    | 26    | 8     |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (60 neighbors) | +0.76 | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | -230        | -286  | N/A   | N/A   |

Nomenclature: See separate page